

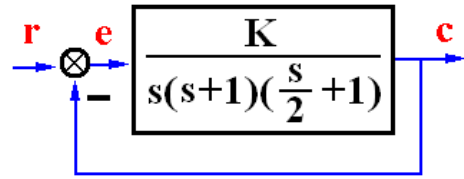
例 2 系统结构图如右图所示

要求：1) 静态速度误差 $e_{ss}^* \leq 0.1$

2) $\gamma^* \geq 40^\circ$

3) $20 \lg h^* \geq 10 \text{ dB}$

试确定校正装置传递函数 $G_c(s) = ?$



解： $G(s) = \frac{K}{s(s+1)(\frac{s}{2}+1)}$ $\begin{cases} \text{开环增益: } K \\ \nu = 1 \end{cases}$

① 依指标确定 K

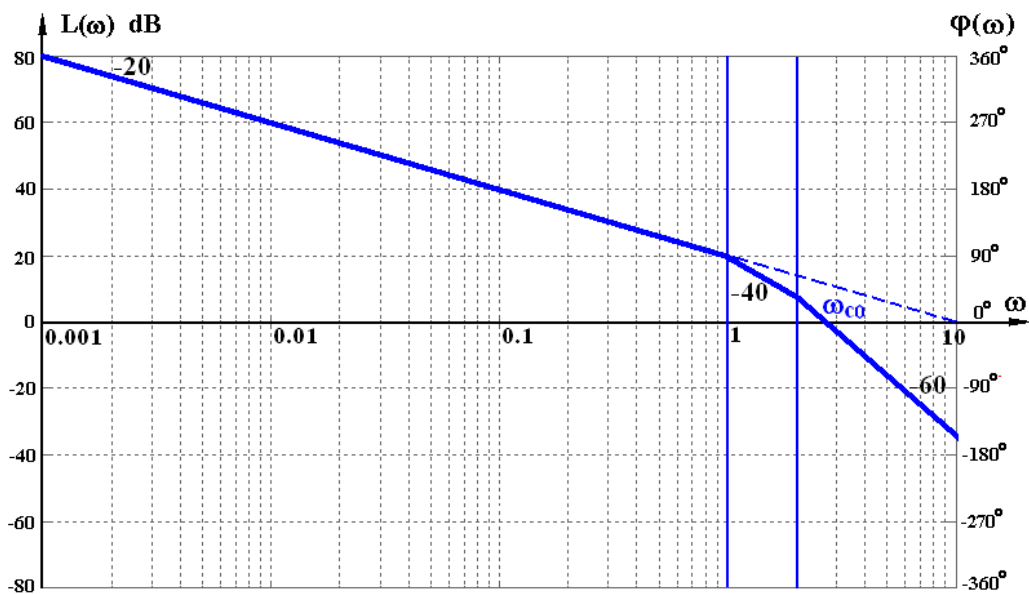
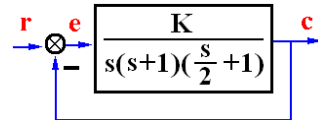
$$e_{ss}^{r=t} = \frac{1}{K} \leq 0.1 \Rightarrow K \geq 10, \text{ 取 } K = 10$$

② $L_0(\omega) \rightarrow \omega_{c0} \rightarrow \gamma_0$

$$|G_0(\omega_{c0})| = \frac{10}{\omega_{c0} \cdot \omega_{c0} \cdot \frac{\omega_{c0}}{2}} = \frac{20}{\omega_{c0}^3} = 1$$

$$\omega_{c0} = \sqrt[3]{20} = 2.7144$$

$$\begin{aligned} \gamma_0 &= 180^\circ - 90^\circ - \arctan \omega_{c0} - \arctan \frac{\omega_{c0}}{2} \\ &= 90^\circ - 68.8^\circ - 53.6^\circ = -33.4^\circ \end{aligned}$$



采用超前校正：

$$\varphi_m = \gamma^* - \gamma_0 + 10^\circ = 40^\circ + 33.4^\circ + 10^\circ = 83.4^\circ > 60^\circ$$

用一级超前校正不能解决问题，试用迟后校正（ ω_c^* 没有要求）。

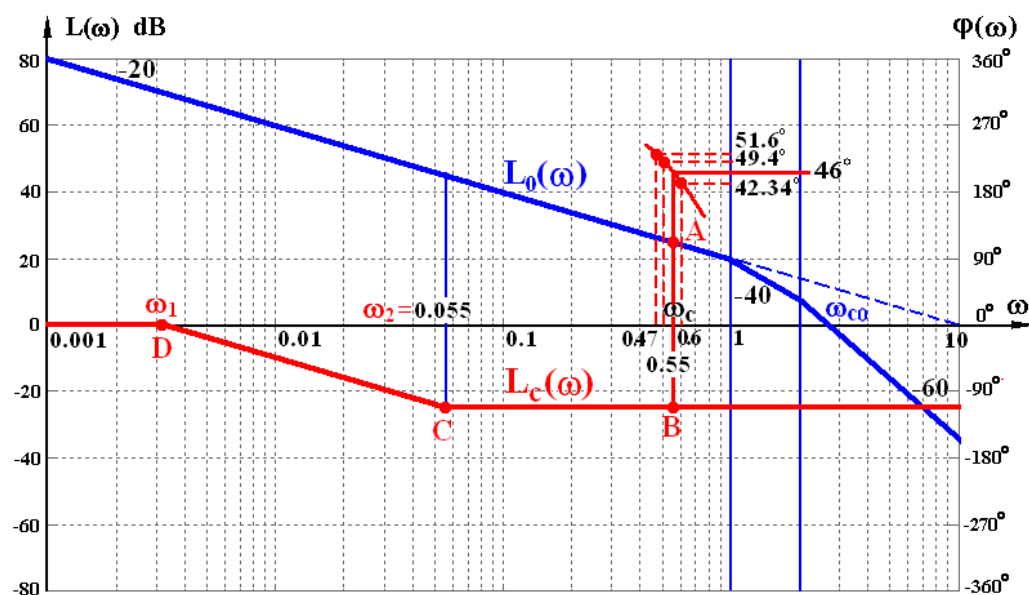
③ 选定 ω_{c1} ，依题，定 $\gamma_c(\omega_{c1}) = \gamma^* + 6^\circ = 40^\circ + 6^\circ = 46^\circ$

作 $\gamma_c(\omega)$ 曲线

ω	$\gamma_c(\omega) = 180^\circ + \angle G_0(j\omega)$
0.6	42.34°
0.5	49.4°
0.47	51.6°

作 $\gamma_c(\omega)$ 曲线， $\gamma_c = 46^\circ \Rightarrow \omega_{c1} = 0.55$

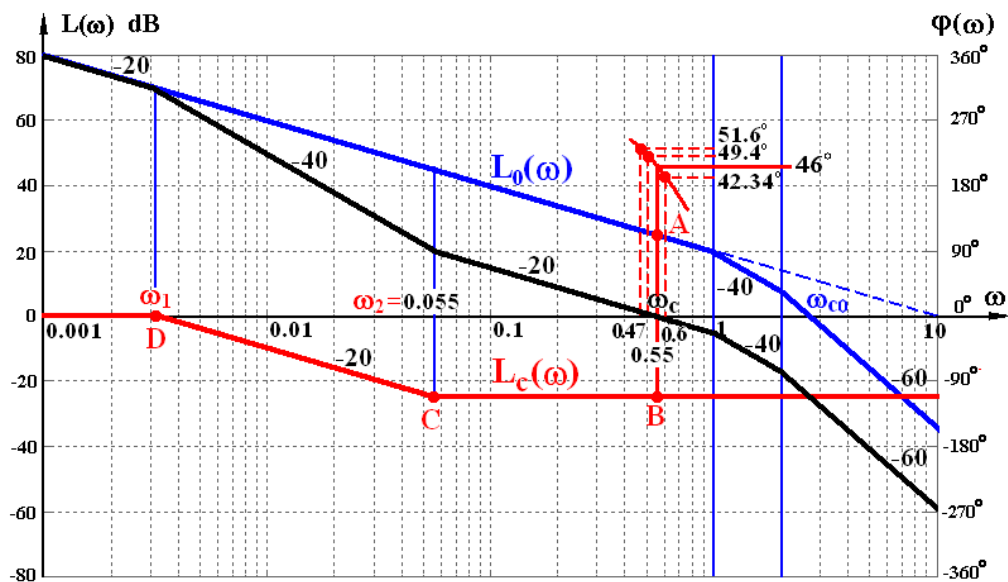
④ 取 $\omega_c = 0.55$ 作图设计：→A, B, C, D，确定 $G_c(s)$



C 点： $\omega_2 = 0.1\omega_c = 0.1 \times 0.55 = 0.055$

D 点： $\frac{10}{\omega_c} = \frac{\omega_2}{\omega_1}$ $\omega_1 = \frac{\omega_c \cdot \omega_2}{10} = \frac{0.55 \times 0.055}{10} = 0.003025$

$$G_c(s) = \frac{\frac{s}{\omega_2} + 1}{\frac{s}{\omega_1} s + 1} = \frac{\frac{s}{0.055} + 1}{\frac{s}{0.003025} + 1} \quad \begin{cases} b = \frac{0.055}{0.003025} = 18.18 \\ T = \frac{1}{\omega_1} = 330.6'' = 5.5(\text{min}) \end{cases}$$



⑤ 验算: $G(s) = G_c(s) \cdot G_0(s) = \frac{10(\frac{s}{0.055} + 1)}{s(s+1)(\frac{s}{2} + 1)(\frac{s}{0.003025} + 1)}$

$$\begin{aligned} \gamma &= 180^\circ + \arctan \frac{0.55}{0.055} - 90^\circ - \arctan 0.55 - \arctan \frac{0.55}{2} - \arctan \frac{0.55}{0.003024} \\ &= 180^\circ + 84.29^\circ - 90^\circ - 28.81^\circ - 15.38^\circ - 89.69^\circ = 40.42^\circ > 40^\circ \end{aligned}$$

检验 h 方法 I:

求 ω_g : 由 $L(\omega)$ 曲线, 应有 $\omega_g \approx \sqrt{2} = 1.414$ 。

试根: $\omega_g = 1.36$

$$|G(\omega_g)| = \frac{10 \sqrt{(\frac{1.36}{0.055})^2 + 1}}{1.36 \sqrt{(1.36)^2 + 1} \sqrt{(\frac{1.36}{2})^2 + 1} \sqrt{(\frac{1.36}{0.003025})^2 + 1}} = 0.1983$$

$$h = \frac{1}{0.1983} = 5.044$$

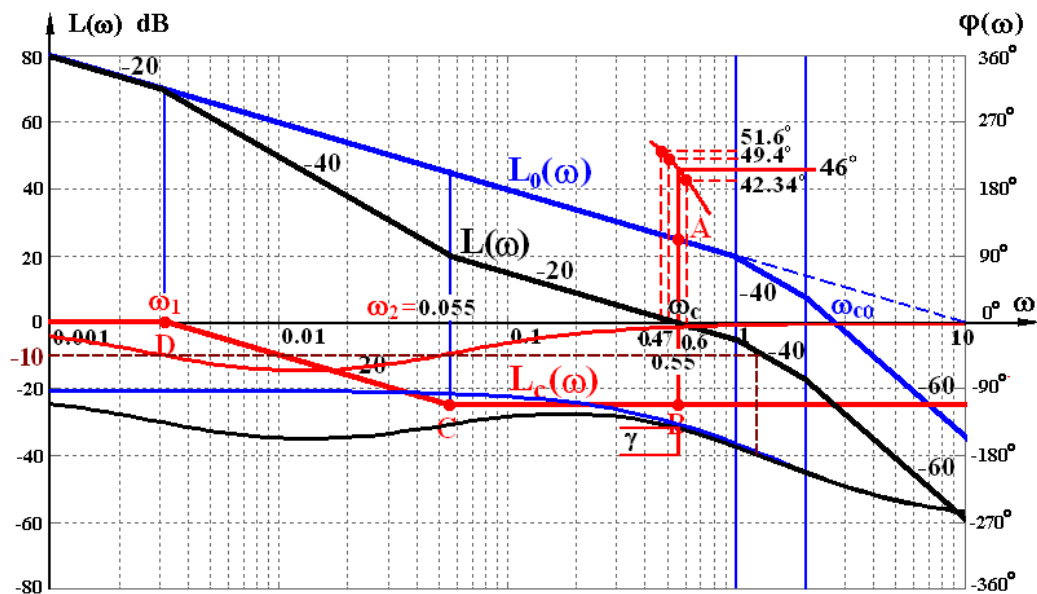
$$20 \lg h = 20 \lg 5.044 = 14.05 \text{ dB} > 10 \text{ dB}$$

检验 h 方法 II:

$$20 \lg K_h = 10 \text{ dB}, \quad K_h = 10^{\frac{10}{20}} = 3.16$$

$$|G(\omega_{g0})| = 3.16 \times \frac{10^{\frac{\omega_{g0}}{0.055}}}{\omega_{g0} \cdot \omega_{g0} \cdot \frac{\omega_{g0}}{0.003025}} = \frac{1.738}{\omega_{g0}^2} = 0$$

$$\begin{cases} \omega_{g0} = \sqrt{1.738} = 1.318 \\ \varphi(\omega_{g0}) = \varphi(1.38) = -178.5^\circ > -180^\circ \end{cases} \quad h > h^* = 10 \text{ dB}$$



迟后校正与超前校正效果的比较:

超前: $\begin{cases} \gamma_c \uparrow, \omega_c \uparrow \rightarrow \text{中频段特性得以改善, 动态性能改善} \\ L(\omega) \text{ 高频段} \uparrow \rightarrow \text{校正后系统抗高频干扰能力} \downarrow \end{cases}$

迟后: $\begin{cases} \gamma \uparrow, \omega_c \downarrow \rightarrow \text{牺牲快速性, 改善均匀性} \\ L(\omega) \text{ 高频段} \downarrow \rightarrow \text{校正后系统抗高频干扰能力} \uparrow \end{cases}$